

World Models

A reading overview and emerging research interest

Group Meeting
September 2026

Outline

1. **World models**

What they model and how their roles differ.

2. **World models for decision making**

Representation, dynamics, and the use of predictions.

3. **My emerging research interest**

How agents retain, revise, and reuse knowledge over time.

The idea of an internal model

- **1943: Craik:** An organism could use a small internal model to anticipate the consequences of possible actions.
- **Control, planning, and RL:** Predicting with a model became a computational procedure, including methods that learn models from experience.
- **2018: Ha and Schmidhuber:** World Models.

The central idea: build a model of the world and predict how it may change.

Craik, *The Nature of Explanation* (1943). Ha and Schmidhuber, *World Models* (2018).

Model-Based Reinforcement Learning (MBRL)

- **Environment model:** In a simple deterministic form,

$$s_{t+1} = f(s_t, a_t), \quad r_t = g(s_t, a_t).$$

Here, f describes how the state changes, and g gives the reward.

- **Model access:** The model may be known in advance, or learned from experience as $\hat{f}$ and $\hat{g}$.
- **Model use:** Predict possible outcomes to plan actions, update value estimates, or train a policy.

MBRL uses an environment model to support learning and decision-making.

What is the difference between WM and MBRL?

	Traditional MBRL	Modern world models
Representation	Often starts with given states or hand-designed features.	Often learns a state/representation from raw observation such as images.
WorldDynamics	Learns the parameter of function f ; Predicts state changes to support planning or policy learning.	Often aims to learn dynamics in latent space, using memory and uncertainty to imagine possible futures.

- MBRL asks how to use a model to learn good actions.
- WM puts more focus on the model itself: what the model should represent and predict
- Difference in representation of the world brings some new problems..

Two functional roles of world models

	Decision-coupled	General-purpose
Main goal	Improve an agent's behavior	Model a range of environments
Model use	Policy learning or planning	Prediction, interaction, or simulation
Emphasis	Information relevant to decisions	Reusable environment dynamics
Examples	Dreamer, TD-MPC series	Genie, Genie 2, GAIA-1

My focus is the decision-coupled side.

Functional axis adapted from Li et al., *A Comprehensive Survey on World Models for Embodied AI*. These roles can overlap.

The decision-coupled workflow

Representation

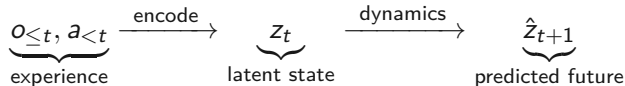
Dynamics

Decision coupling

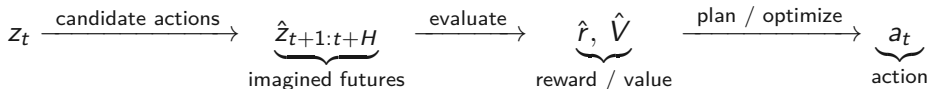
Deployment



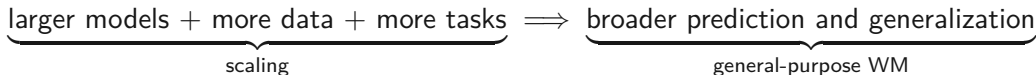
1. Learn an internal predictive model



2. Couple prediction with decision-making



3. Scale the model and data



Representation: what should the model retain?



- **Reconstruction-based learning:** Retain features that help reconstruct observations and predict futures (Dreamer).
- **Task-oriented latent learning:** Learn features for latent, reward, and value prediction (TD-MPC series).
- **Token representations:** Encode inputs as feature sequences for temporal or multimodal modeling.

Does the representation preserve the distinctions needed for control?

Hafner et al., Dreamer; Hansen et al., TD-MPC2; Li et al., survey.

Dynamics: how does the world change?

Representation

Dynamics

Decision coupling

Deployment



$$\hat{z}_{t+1} = f_{\theta}(z_t, a_t) \quad \text{or} \quad \hat{z}_{t+1} \sim p_{\theta}(\cdot \mid z_t, a_t)$$

- **Prediction horizon:** small local errors can accumulate during rollouts.
- **Uncertainty:** several futures may be consistent with the available information.
- **True dynamics:** are we learning the dynamics, or merely predictive patterns?

If the rules of the real world change, how can the model notice that its old predictions are no longer valid?

Representative methods: Dreamer and TD-MPC2.

Decision coupling: policy learning and planning

Representation

Dynamics

Decision coupling

Deployment



Dreamer

TD-MPC series

Imagination	Train a policy in latent space	Evaluate candidate action sequences
Execution	Select actions with the policy	Replan after each observation
Key question	Does imagined learning improve real behavior?	Do predicted returns rank plans correctly?

The goal is to turn predictions into better behavior in the real environment.

Hafner et al., Dreamer; Hansen et al., TD-MPC; Hansen et al., TD-MPC2.

Scaling, generalization, and deployment



- **Scaling:** Share knowledge across tasks (TD-MPC2), or connect video pretraining to robot prediction (V-JEPA 2).
- **Generalization:** Reuse knowledge under new goals, appearances, dynamics, or embodiments.
- **Deployment:** Balance decision quality with latency, noise, and the cost of real interaction.

Success on many training tasks leaves open what transfers to unseen conditions.

Hansen et al., TD-MPC2; Assran et al., V-JEPA 2.

My interest: knowledge that accumulates over time

How can an agent retain, revise, and reuse knowledge across experience?

- **Reuse across tasks**

Can related tasks share representations, dynamics, skills, or experience?

- **Adapt when experience conflicts**

Does an error mean insufficient learning, changed dynamics, or a representation that misses an important distinction?

- **Preserve while learning**

What should change, what should remain, and how can new abilities be acquired without damaging useful old knowledge?

Learning over time is not only about fitting new data: the agent must decide what to reuse, what to revise, and what to preserve.

Continual learning as a research framework

Continual learning studies how a system learns from sequential experience while retaining and using prior knowledge.

- **Retention:** Preserve useful knowledge and earlier capabilities.
- **Adaptation:** Acquire new knowledge when conditions or demands change.
- **Transfer:** Let previous learning improve subsequent learning.

Background reading: De Lange et al., *A Continual Learning Survey*.

Possible settings for the broader question

World-model agents

Study learned representations and predictive dynamics.

Observe how model changes affect control and planning.

LLM agents

Study persistent memory, tool use, and workflows.

Observe how accumulated experience affects later tasks.

Shared question

Which knowledge should the system preserve, revise, or reuse as experience accumulates?

World models and LLM agents may offer different perspectives on this question, and the most suitable entry point can be further discussed depending on the specific mechanism of interest.